PICOCTF WRITEUP

Challenge Name

Share Secret

Level: Easy
Points: 100

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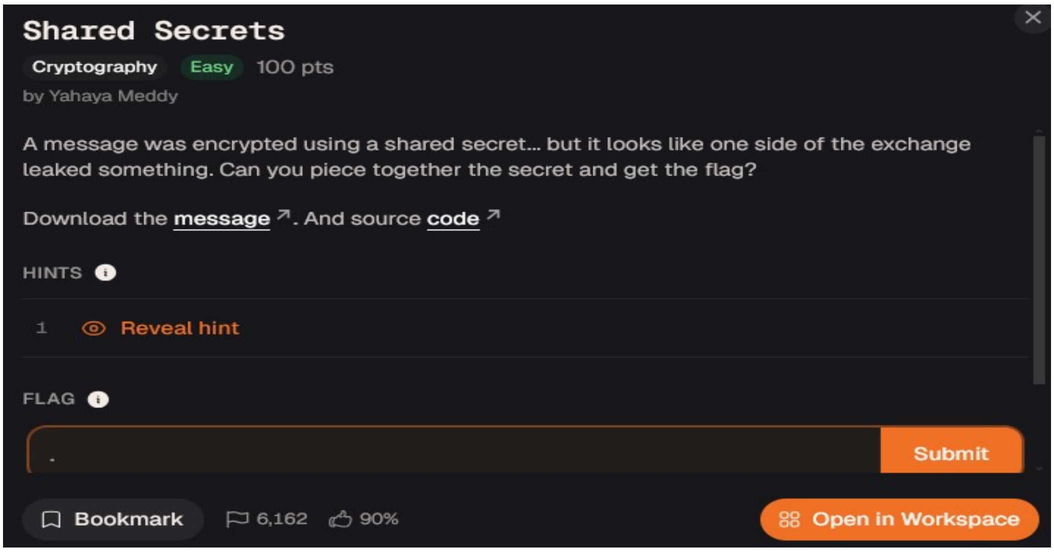


Challenge Description

A message was encrypted using a shared secret... but it looks like one side of the exchange leaked something. Can you piece together the secret and get the flag?

Step 1

I downloaded two files:

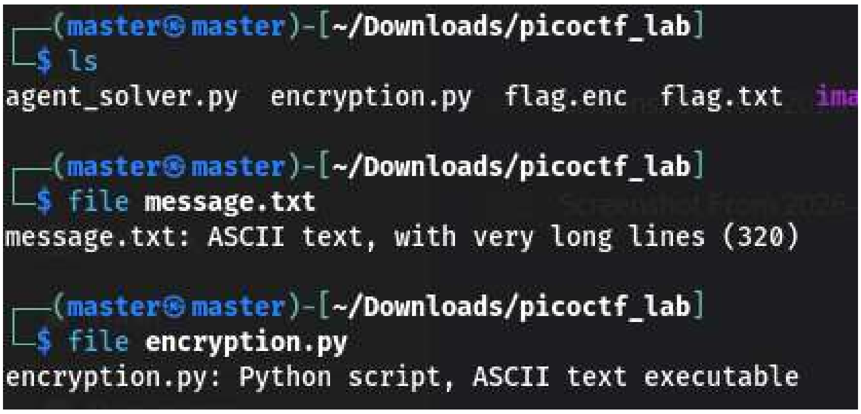
```
message.txt

encryption.py
```

Step 2

I checked the files using the `file` command.

```
file message.txt
file encryption.py
```



Then I read the content of the message file.

```
cat message.txt
```

Output:

```
g = 2
p =
21320040263031091389604193705821048453829391592318162736206967012946302
84403386465532863373769144582236949856392667504369986764250039517010152
81593114087468259049658291367942212550000829687053424636659474573599325
69887147233770302933744967707447704105478174116064585960580227097589939
68788819254528632650187419776389

A =
11419333687496515472851065847700229465985562945322554840182340712790172
84493261458304303078441814350280055235938839850330392924394827971935746
68475672410615478352986620668636434239178081957738980683502215418709666
52299782754137487140492437407367794890184382784787080685735113384518733
71736513057724525994219285563327

b =
73862321156174637262417094403087048594497818743384458341232576077734138
08671041124877432868121123946611876129024302142022027663780163759048916
04068375412515990443266634874912487423407082997026224391496763715710675
```

```
14516396771533967528368942450713740842514358149618644686972388555786539
0662886201409684909929549168034

enc = 928b818da1b6a499868abd91d18190d196bdd1d08781d0d4d5db9f
```

I found different values including Alice's public key (`A`) and Bob's private key (`b`).

Step 3

I created a Python script to combine Alice's and Bob's values and recover the shared secret used for encryption.

```
import hashlib

g = 2
p =
21320040263031091389604193705821048453829391592318162736206967012946302
84403386465532863373769144582236949856392667504369986764250039517010152
81593114087468259049658291367942212550000829687053424636659474573599325
69887147233770302933744967707447704105478174116064585960580227097589939
68788819254528632650187419776389

A =
11419333687496515472851065847700229465985562945322554840182340712790172
84493261458304303078441814350280055235938839850330392924394827971935746
68475672410615478352986620668636434239178081957738980683502215418709666
52299782754137487140492437407367794890184382784787080685735113384518733
71736513057724525994219285563327

b =
73862321156174637262417094403087048594497818743384458341232576077734138
08671041124877432868121123946611876129024302142022027663780163759048916
04068375412515990443266634874912487423407082997026224391496763715710675
14516396771533967528368942450713740842514358149618644686972388555786539
0662886201409684909929549168034

enc_hex = "928b818da1b6a499868abd91d18190d196bdd1d08781d0d4d5db9f"

shared_secret = pow(A, b, p)

secret_bytes = str(shared_secret).encode()
key = hashlib.sha256(secret_bytes).digest()

ciphertext = bytes.fromhex(enc_hex)

flag = "".join(chr(byte ^ 0xe2) for byte in ciphertext)

print(flag)
```

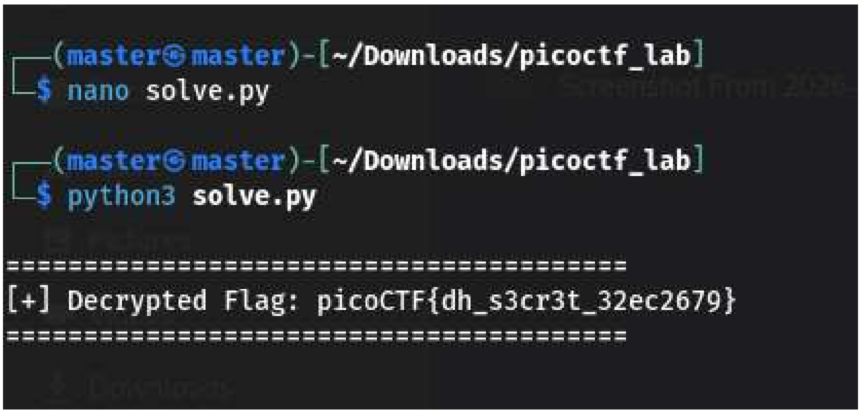
Step 4

I created a file called:

```
nano solve.py
```

Then I saved the code and executed it.

```
python3 solve.py
```



Flag

```
picoCTF{dh_s3cr3t_32ec2679}
```

THANK YOU ALL
Happy h4cking